

$n = \text{power}$

1. Differentiation: at Instantaneous

• Change in something with respect to something.

Differentiation $\frac{dx}{dt}$ = change in "x" with due to "t".

• 6 Laws.

$$\left[1) \frac{d(x)^n}{dx} = n(x^{n-1}) \right]$$

$$\text{Ex: } \frac{dx^5}{dx} = 5x^4$$

$$\text{Ex: } \frac{dy^3}{dy} = 3y^2$$

$$\begin{aligned} \text{Ex: } \frac{dy^{-3}}{dy} &= -3y^{-3-1} \\ &= -3y^{-4} \end{aligned}$$

$$\begin{aligned} \frac{dm^{\frac{3}{2}}}{dm} &= \frac{3}{2} m^{\frac{3}{2}-1} \\ &= \frac{1}{2} \\ &= \frac{3}{2} m^{\frac{1}{2}} \end{aligned}$$

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$$2. \frac{dx}{dx} = 1$$

$$3. \frac{d(\text{constant})}{dx} = 0$$

$$\text{Ex: } \frac{d(3)}{dx} = 0$$

$$\text{Ex: } \frac{d(\pi)}{dx} = 0$$

$$\text{Ex: } \frac{d(\sin 30^\circ)}{dx} = 0$$

$$\frac{dx^n}{dx} = n(x^{n-1})$$

$$\left(4. \frac{d(\text{constant} \times x^n)}{dx} \Rightarrow \text{constant} \left(\frac{dx^n}{dx} \right) \right)$$

$$\text{Ex: } \frac{d(3x^3)}{dx} \Rightarrow 3 \left(\frac{dx^3}{dx} \right)$$

$$= 3 \times 3x^2$$

$$= 9x^2$$

$$\text{Ex: } \frac{d(4y^5)}{dy} = 4 \times \left(\frac{dy^5}{dy} \right)$$

$$= 4 \times 5y^4$$

$$= 20y^4$$

$$\text{Ex: } \frac{d5x^{-2}}{dx} = 5 \left(\frac{dx^{-2}}{dx} \right)$$

$$= 5x^{-2 \times -1}$$

$$= 5x^{-2} - 2x^{-3}$$

$$= -10x^{-3}$$

$$4. \frac{d(m \pm n)}{dx} = \frac{dm}{dx} \pm \frac{dn}{dx}$$

$$\text{Ex: } \frac{d(x^3 + x)}{dx} = \frac{dx^3}{dx} + \frac{dx}{dx}$$

$$\Rightarrow 3x^2 + 1$$

$$\text{Ex: } \frac{d(y^2 + y - y^2)}{dy} = \frac{dy^2}{dy} + \frac{dy}{dy} - \frac{dy^2}{dy}$$

$$= 2y^1 + 1 - 2y^1$$

5. Product Rule:

$$\frac{d(m \times n)}{dx} = \frac{dm}{dx} \times n + \frac{dn}{dx} \times m$$

6. Some Variables:-

$$\frac{d(\sin x)}{dx} = \cos x$$

$$\frac{d(\cos x)}{dx} = -\sin x$$

Ex:- $\frac{d(x^2 \sin x)}{dx} = \frac{dx^2}{dx} \times \sin x + \frac{d(\sin x)}{dx} \times x^2$

$$\Rightarrow 2x \sin x + \cos x \times x^2$$

Ex:- $\frac{d(y^4 \cos y)}{dy} \Rightarrow \frac{dy^4}{dy} \times \cos y + \frac{d(\cos y)}{dy} \times y^4$

$$= 4y^3 \times \cos y + (-\sin y) \times y^4$$

6. Dividend Rule:-

$$\frac{d(m/n)}{dx} = \frac{\frac{dm}{dx} \times n - \frac{dn}{dx} \cdot m}{n^2}$$

Ex. $\frac{d(x^2/\sin x)}{dx}$

$$\frac{\frac{dx^2}{dx} \times \sin x - \frac{d \sin x}{dx} \times x^2}{\sin x^2}$$

$$\Rightarrow \frac{2x \cdot \sin x - \cos x \cdot x^2}{(\sin x)^2}$$

$$Q \quad \frac{d\left(\frac{\cos x}{x^5}\right)}{dx}$$

$$= \frac{d(\cos x)}{dx} \cdot \frac{1}{x^5} - \frac{d x^5}{dx} \cdot \frac{\cos x}{x^5}$$

$$= \frac{-\sin x \cdot x^5 - 5x^4 \cdot \cos x}{x^{10}}$$

~~x x^2~~